

1 MLE Communication

Purpose

You use the MES Link Enabling (MLE) communication to exchange data between the MES and other systems, e.g. ERP or warehouse management systems. MES Link Enabling provides a framework that can be used by the applications (application interfaces).

Integration

The MLE framework is used by many application interfaces.

Basics

HYDRA provides the *MES Link Enabling communication*, in the following referred to as *MLE*, for the communication with ERP systems in general and SAP R/3 or ECC in particular.

MLE is a program environment enabling the data transfer with external systems. For this purpose, we use:

- file servers and file clients for the data exchange on file level and
- RFC servers and RFC clients for the data exchange with SAP

The MLE inbound dispatcher for HYDRA inbound processing is used in any case, irrespective of the external system data is exchanged with.

IDoc (Intermediate Document)

Use IDocs (intermediate documents) to exchange data:

- within an ERP system
- between several ERP systems or
- between ERP systems and third-party systems.

IDocs are containers to exchange data between systems. IDocs may be flat or have multi-level hierarchies. IDocs use data segments to summarize data in logical units. IDocs combine data contents of the same nature and structure. Therefore, you can transfer several of these “clusters” within one file. Although each IDoc corresponds to a defined data type/structure, the format does not depend on the content or type of content.

In general, IDocs consist of a control record, multiple data segments and the status record. There is exactly one control record for each IDoc. The control record includes all pieces of information required for sending and processing the IDoc. This information includes, for example:

- the message type
- the IDoc type
- the sending and receiving (logical) system. The control record is like an envelope stating address details.

The data segments of the IDoc include the data. Each segment consists of preliminary information specifying the data structure. The payload (user data) itself is stored in a uniform, unstructured field. Segments can also be structured in hierarchies.

Different processing steps are executed while an IDoc is being processed. Another status record is added for each step. You can use these records to keep track of the processing steps and their results.

RFC technology

The Remote Function Call (RFC) constitutes the basis for the data exchange between SAP R/3 or ECC and subsystems. The *remote function call* allows you to call SAP functions remotely from other systems. The *transactional Remote Function Call* (tRFC) is an upgrade of this technology.

The *transactional RFC* provides independence of other systems. The basis for this procedure is the dispatching system's obligation (RFC client) to call the receiving system over and over again, provided that communication could not be established or in case of an interruption. However, this only repeats until the transaction has been transferred successfully.

A worldwide unique transaction number (TID) is allocated for each transaction in order to guarantee consistent data retention even if communication is interrupted.

MLE RFC server

An RFC server logs on to an SAP gateway and waits for data. When data arrives, the server receives and stores the data in a database or file. Then, the server waits for data again.

There are different modes to log on an RFC server to an SAP gateway. The *Registering Mode* is used for mySAP communication. In this case, the RFC server registers its functions at the SAP gateway. Therefore, these functions can be called in the SAP environment.

Different older communication models use other modes which, however, are not relevant to HYDRA.

The MLE-ALE server (hyalesrv.exe) takes over the RFC server functions, when it comes to the HYDRA MLE communication.

MLE RFC client

An RFC client transfers data from HYDRA to SAP R/3 or ECC. To do so, the RFC client logs on to an SAP system and calls a specific function module. This can be a synchronous or asynchronous call.

The MLE-RFC client takes over the outbound communication with R/3. The data segments created in HYDRA are transferred to the respective target system according to distribution specifications.

An upload to R/3 can be triggered in two ways.

- Triggered by an upload request from R/3:

In this case, the RFC client has to be entered as processing program for the upload request in the distribution model. The segment names of the upload IDoc must be entered as its parameters.

- Cyclical upload

You can use the HYDRA Scheduler if you want the upload to be triggered by HYDRA.

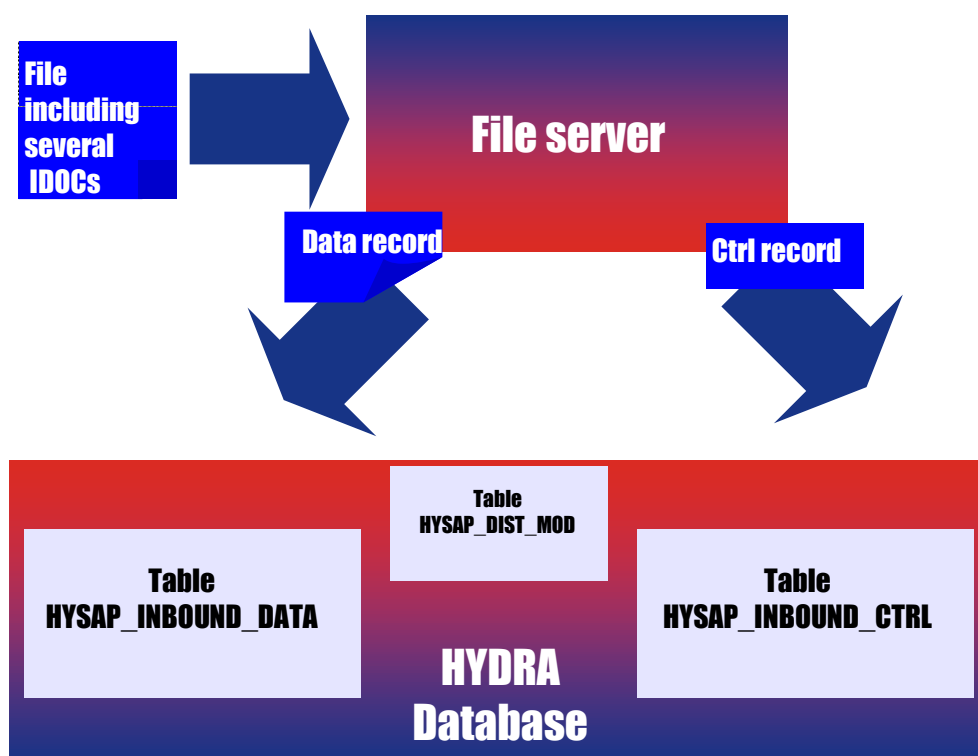
MLE file server

Besides the data exchange via remote function call, data can also be exchanged using file transfer. In this case, the file server monitors a specific directory to identify new files and transfers these new files to HYDRA. The file client, in turn, converts HYDRA uploads into files and archives them also in a specified directory.

The files have to be named like the message types defined in the HYDRA MLE distribution model. HYDRA inbound processing may be controlled by the file extension. The following file extensions are possible:

- "APP" Creating/modifying data
- "UPD" Replacing the data existing in HYDRA by the newly transferred data (initial download).
- "DEL" Deleting data
- "DAT" The extension is used for all other files. The message function "DAT" is not transferred to the IDoc control record.

If files assigned to the names of the distribution model and the above-mentioned extensions are found, these files are copied to the working directory. This working directory is also specified in the configuration of the HYDRA MLE file server. The system then transfers the data records existing in this working directory to the inbound tables of the HYDRA MLE interface.



MLE file client

HYDRA provides data (e.g. confirmations/uploads of operations) in the interface. According to the respective segment type, HYDRA files the data as such segments.

This data is no longer interpreted for the actual communication layer. These segments are summarized in IDocs and provided with connection details through the configuration of outbound processing. A file client then makes these segments available in a specified directory where they can be processed by the ERP system.

The HYDRA MLE file client performs outbound processing on the basis of ASCII files. If a logical system created as file interface is assigned to a message type, the data is archived in a file in a specific directory. This directory is specified by the configuration of the HYDRA MLE file client.

HYDRA controls the upload or writing of the files into the specified directory at regular intervals.

MLE inbound dispatcher

The HYDRA MLE Inbound Dispatcher organizes inbound processing in HYDRA.

The *HYDRA MLE Inbound Dispatcher*:

- monitors inbound transactions;
- uses the message type (from the MLE distribution model) to identify the corresponding processing routine (program) to transfer data to HYDRA;
- starts this processing routine. Inbound transactions are processed according to the sequence specified by the ERP system. Consequently, a transaction can only be processed, once the previous transaction has been completed.

The following diagram shows the processing steps of the dispatcher when processing inbound transactions:



The system generates log and error files for the data transferred to HYDRA.

Logical systems

Each system landscape consisting of test and production system(s), is represented as logical system in HYDRA. HYDRA's configuration for inbound and outbound processing refers to this logical system.

Please note for SAP R/3 or ECC:

Third-party systems must be configured as logical system with a unique name in R/3 and/or ECC in order for SAP R/3 and/or ECC to be able to communicate with these external systems.

This name is used as the "address" for sending IDocs. In return, the R/3 or ECC system is also configured as logical system.

Procedure for transferring data to HYDRA

An ERP interface program generates the data with the defined data structures and stores this data in a transfer file. This file can then be transferred to HYDRA.

These transfer files are to be provided in the HYDRA subdirectory `./inf_int/interf` (standard system). You can find this directory in the HYDRA directory or in case of multiple system environments in the relevant system directory.

You have to implement a handshake logic to transfer the files between the ERP system and HYDRA. This logic prevents any data losses by "overwriting" transfer files.

Note the following in order to ensure reliable processing:

The file to be provided by the ERP system must not exist under the documented name until HYDRA releases it for transfer and processing.

When transferring the file from the source system to the HYDRA server, you have to use another name for the transfer file (a name **other than** the documented name). Rename the file and use the specified file name, once it has been transferred. To rename the file:

- Use the REN command for Windows systems
- Use the mv command for UNIX/LINUX. The file extensions ".APP", ".UPD", ".DEL" and ".DAT" are reserved for HYDRA. The ".TRF" extension is recommended.

If the transfer directory already includes a file, the system has to wait until HYDRA has taken over this file before transferring a new file.

If the ERP system creates the file directly on the HYDRA server, you have to ensure that the file does not exist under the documented name when it is created, written or appended. However, MPDV does not recommend this procedure in general.

To prevent the file size from growing endlessly, the ERP system should interrupt the write process at 50 Mbyte and wait for the file server to collect the data. A new file can be written, once the file server has removed the file. This is especially important for the initial download.

The file port processes the data included in the interface directory as follows:

When starting the service / process (hyalesrv.exe/out), the system identifies the message types to be integrated and sorts them according to the defined priority.

The priority is digital ("0" = no priority / "1" high priority) and can be defined in the HYDRA distribution model.

In the interface directory a HYDRA process searches for the files as follows within one message type:

1. DEL / del
2. APP / app
3. UPD / upd
4. DAT / dat

Procedure for transferring data from HYDRA

An ERP interface program prepares the data structures of the files transferred by HYDRA.

You have to implement a handshake logic to transfer the files between the ERP and HYDRA. This logic prevents any data losses by "overwriting" transfer files.

These transfer files are provided in the HYDRA subdirectory ./inf_int/interf (standard system). You can find this directory in the HYDRA directory or in case of multiple system environments in the relevant system directory.

Use the following processing method to safely process the files:

- Rename the upload file using:
 - the REN command with NT and
 - the mv command with UNIX.
- Transfer the renamed upload file to the ERP environment.

Notes

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| <ul style="list-style-type: none">- You must <u>NOT</u> use a copy command in this step. |
|--|

- As long as HYDRA is processing the file, the file does **not** exist under the specified name. This ensures that the higher-level system can only access the file, once HYDRA has finished processing (secure handshake).
- The file extensions ".APP", ".UPD", ".DEL" and ".DAT" are reserved for HYDRA. The ".TRF" extension is recommended.

Automated interfacing

MPDV recommends using a gateway including corresponding control software to allow for the HYDRA server and the customer's server to be connected automatically. This gateway controls the data exchange between the two systems in both directions. As an alternative, you can use any other procedure that ensures a secure file transfer, in particular the connection of both systems using a network file system.

Notes on the file format/codepage

The following conventions apply for the codepage/character set:

MES-Weaver 3.0

Each data record included in the file has to be completed by 'CR' (U+000D) and 'LF' (U+000A) for Windows and 'LF' (U+000A) for Unix.

HYDRA expects the file to be in the UTF-8 format and HYDRA also uses this format for uploads. On request, you can also transfer files in the file format that was used until MW 2.0.